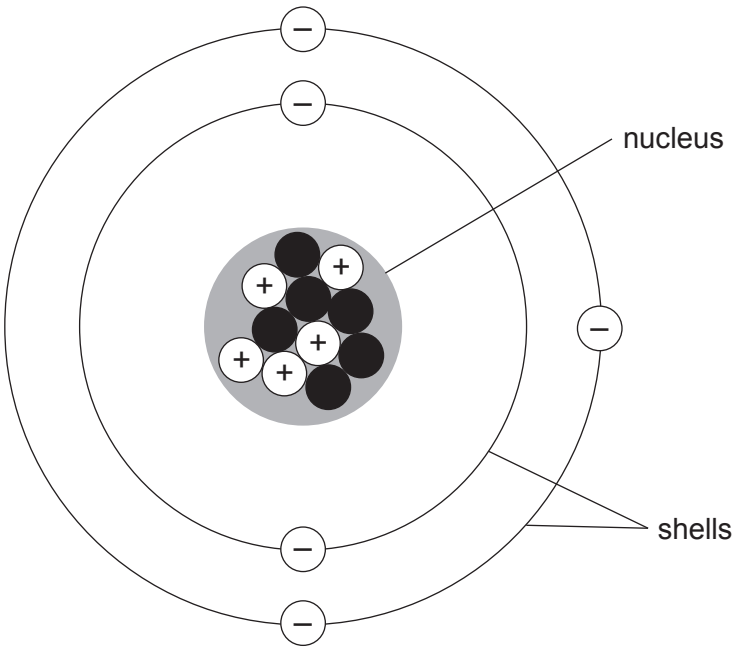


2. (a) Atoms contain particles called protons, neutrons and electrons. The diagram shows a model of an atom of boron.



State whether the statements in the table below are **true** or **false**. [4]

Statement	True or false?
Boron atoms contain the same number of protons and electrons	
The particles found in the shells are called electrons	
The nucleus contains five neutrons	
The electronic structure of boron is 3,2	

- (b) The formula of boron trioxide is B_2O_3 .

Calculate the relative formula mass (M_r) of boron trioxide.

[2]

$$A_r(B) = 11 \quad A_r(O) = 16$$

Relative formula mass =

- (c) The diagrams below represent atoms of boron and fluorine.



boron, B

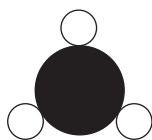


fluorine, F

Boron trifluoride has the formula BF_3 .

Choose the **letter** of the diagram that represents a molecule of boron trifluoride.

[1]



A



B



C

Letter

- (d) Magnesium fluoride contains the ions Mg^{2+} and F^- .

Underline the correct formula for magnesium fluoride.

[1]

